

Optical Simulation of Scintillator Bars EJ-204 vs EJ-230 for the SHiP Timing Detector

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Scintillator bar (GEANT4):

- Active length: 1400 mm (± 700 mm along x)
- Coating: reflective Mylar
- $v_{\text{eff}} \approx 277$ mm/ns (nominal axial average)

→ ToF onset measures $v_{\text{eff}}^{\text{onset}} \approx 292$ mm/ns (+5%): set by the most direct rays, not the mean path (see QA-1c)

- Two materials compared in parallel:
EJ-204 vs **EJ-230**

End SiPMs (primary timing):

- 16 SiPMs total (IDs 0–15)
- END_LEFT: IDs 0–7 at $x = -700$ mm
- END_RIGHT: IDs 8–15 at $x = +700$ mm
- Timing estimator: **SUM4** (min over 4-neighbor clusters)

Top SiPMs (veto / tracking):

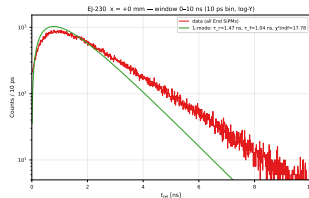
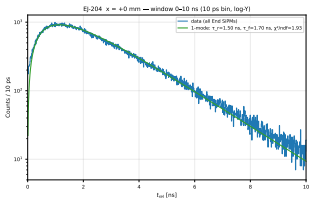
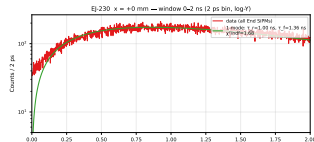
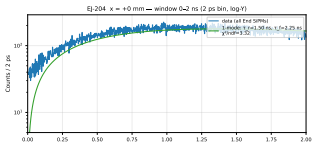
- 70 SiPMs, single row at $Y = +30.25$ mm
- TOP_LEFT: IDs 16–50 / TOP_RIGHT: 51–85
- Pitch: 20 mm in X ; central gap 24 mm

GEANT4 statistics:

Dataset	Events	Positions
EJ-204 End	2000/pos	31
EJ-230 End	2000/pos	31
EJ-204 EndTop	2000/pos	31

Positions: $x = 0, \pm 50, \dots, \pm 690$ mm

F1: Relative arrival time t_{rel} — central position $x = 0$ EJ-204 (left) vs EJ-230 (right) — 1-mode fit: $N(1 - e^{-t/\tau_r})e^{-t/\tau_f}$



EJ-204: $x=+0$ mm, $t_{\text{cut}}=t_{\text{prompt}} \rightarrow \text{PH}(t_{\text{cut}})$ per event

EJ-230: $x=+0$ mm, $t_{\text{cut}}=t_{\text{prompt}} \rightarrow \text{PH}(t_{\text{cut}})$ per event

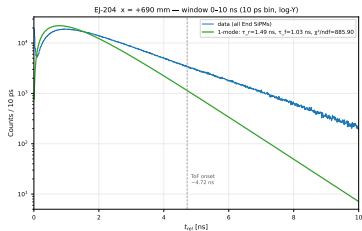
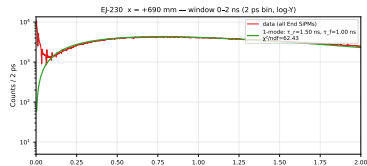
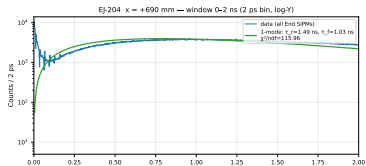
$t_{\text{rel}} = t_{\text{photon}} - \min(t_{\text{End}})$ per event. EJ-204: $\tau_r = 1.50$ ns, $\tau_f = 1.70$ ns ($\chi^2/\text{ndf} = 1.93$). EJ-230: $\tau_r = 1.47$ ns, $\tau_f = 1.04$ ns ($\chi^2/\text{ndf} = 17.8$).

Fitted τ describe the arrival-time profile (emission $\otimes$ propagation), NOT the datasheet emission constants (EJ-204: $\tau_r^{(0)} = 0.7$ ns, $\tau_d^{(0)} = 1.8$ ns).

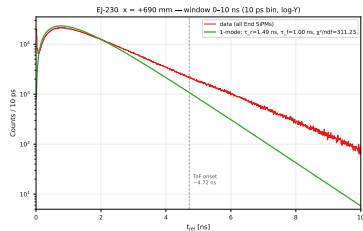
EJ-230 tail at $x = 0$: **cannot** be ToF (ends equidistant); origin undetermined — slow mode vs reflections.

F1: Relative arrival time t_{rel} — edge position $x = +690$ mm

Second component at ~ 4.7 ns visible in both materials



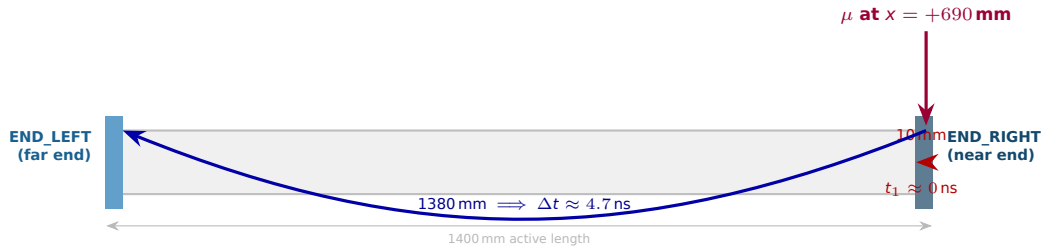
EJ-204 $x = +690$ mm. $t_{\text{off}} = t_{\text{offset}} - \text{min}(t_{\text{off}})$ per event.
Tail 3-10 ns: ~90% scattered near-end photons + ~2% far-end ToF (QA-1c onset=4.72ns).



EJ-230 $x = +690$ mm. $t_{\text{off}} = t_{\text{offset}} - \text{min}(t_{\text{off}})$ per event.
Tail 3-10 ns: ~90% scattered near-end photons + ~2% far-end ToF (QA-1c onset=4.72ns).

1-mode fit fails at $x = +690$: $\chi^2/\text{ndf} = 886$ (EJ-204), 721 (EJ-230). Tail 3-8 ns: **not** slow scintillation — see ToF diagram below and QA-1c.

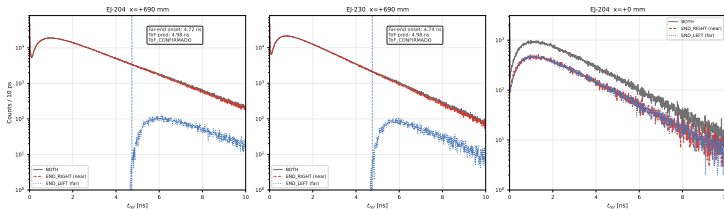
Why a second component appears at the edge — geometric time-of-flight



The late bump is the *same* photons reaching the *far end* — not slow scintillation.

QA-1c: Second F1 component confirmed as geometric ToF

QA-1c — L-rel per end | Grey=BOTH Red=END_RIGHT (near) Blue=END_LEFT (far)
Far end appears entirely at $t_{\text{rel}} > 4$ ns (ToF confirmed, onset 4.72 ns vs 4.98 ns predicted)



ToF prediction ($x = +690$ mm):

$$\Delta t_{\text{pred}} = \frac{1380 \text{ mm}}{277 \text{ mm/ns}} \approx 4.98 \text{ ns}$$

Measured onset (far end):

- EJ-204: 4.72 ns (−5.3%)
- EJ-230: 4.74 ns (−4.9%)

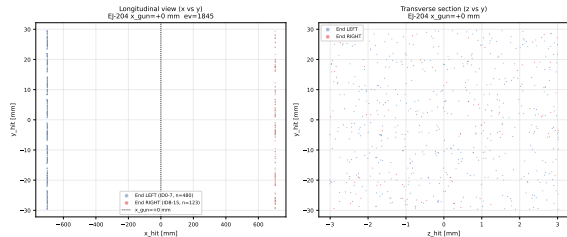
Verdict:

ToF_CONFIRMED

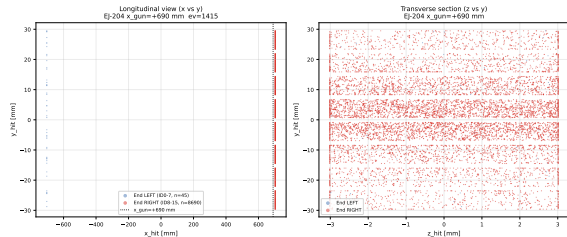
The −5% offset reflects $v_{\text{eff}}^{\text{onset}} \approx 292 \text{ mm/ns}$ — the fastest (most direct) optical rays arrive first, pulling the onset earlier than the mean-path prediction.

F3: Spatial distribution of photon impacts on SiPMs

Single representative event — EJ-204 (EJ-230 qualitatively identical)



F3 — Photon hits on SiPMs (NOT volume tracks), event 1845 (603 hits). EJ-204 event display reconstructed from hits (not re-simulation).



F3 — Photon hits on SiPMs (NOT volume tracks), event 1415 (8735 hits). EJ-204 event display reconstructed from hits (not re-simulation).

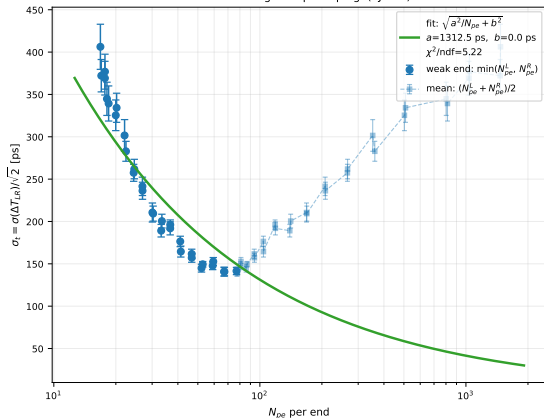
$x_{\text{gun}} = 0 \text{ mm}$ — symmetric L/R distribution

$x_{\text{gun}} = +690 \text{ mm}$ — collapse toward END_RIGHT

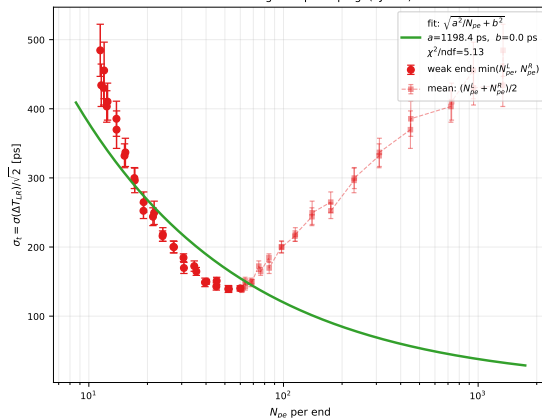
Note: Points are *photon impacts on the SiPM window* (NOT trajectories inside the scintillator volume). Proxy of optical propagation.
Reconstructed from x_{mm}/y_{mm}/z_{mm} ROOT branches — no re-simulation.

F2: σ_t vs N_{pe} (END SiPMs) — pure Poisson, more light always helps

F2: timing resolution vs photon yield
where more light stops helping (EJ-204)



F2: timing resolution vs photon yield
where more light stops helping (EJ-230)



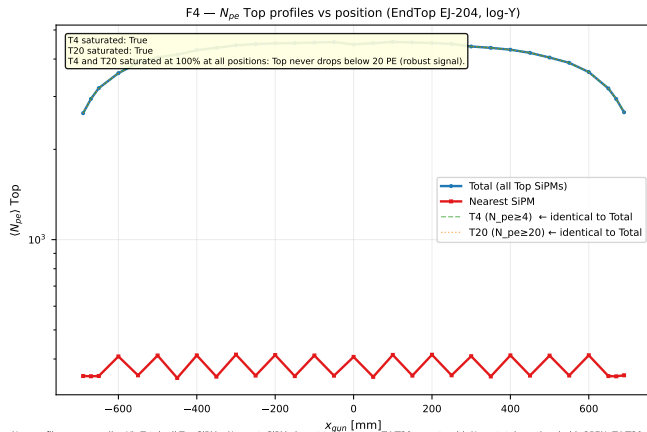
END SiPMs (IDs 0-15, timing); each point = one muon position; $x = N_{pe}^{\text{weak}}$ (far-end photons); $y = \sigma_t$ (same as F5). For Top SiPM light profiles, see F4.

EJ-204: $a = 1312 \pm 24$ ps, $b = 0$, $\chi^2/ndf = 5.22$ **EJ-230:** $a = 1198 \pm 23$ ps, $b = 0$, $\chi^2/ndf = 5.13$

Pure Poisson ($\sigma_t \propto 1/\sqrt{N_{pe}}$, $b = 0$): no saturation in $N_{pe}^{\text{weak}} \in [11, 77]$. **Edge timing is limited by scarce far-end light, not by propagation spread. Adding light always improves edge timing.**

F2&F7 consistent: optical-path dispersion EXISTS (F7 RMS > floor) but *subdominant* to Poisson statistics in SUM4 σ_t ($b \approx 0$).

F4: Top SiPM light yield vs position — not a timing quantity (EndTop EJ-204)



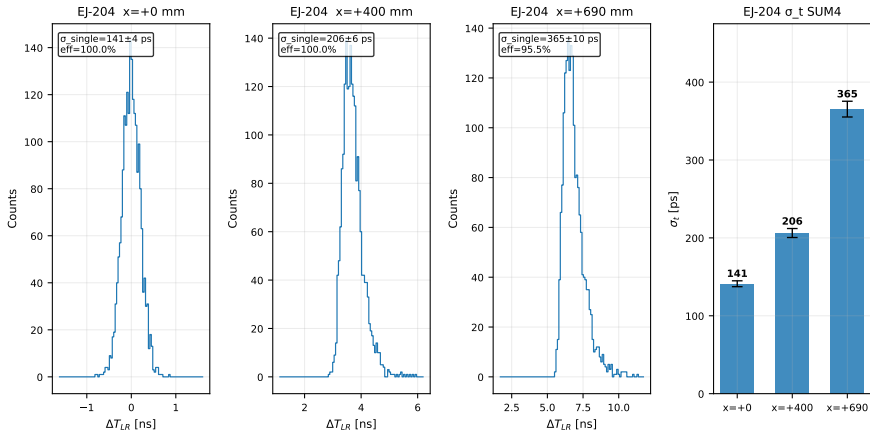
EJ-204 (EndTop) Top Npe profiles vs x_{gun} (log-Y). Total: all Top SiPMs. Nearest: SiPM closest to the muon. T4/T20: events with $N_{pe_total} \geq$ threshold. OPEN: T4/T20 definition pending Gerardo.

- **TOP SiPMs** (IDs 16–85): Y-axis = light yield (*not* timing; for σ_t see F2/F5). T4 and T20 **saturated at 100%** at all positions: Top never falls below 20 PE — robust signal across the full range.
- “Nearest” curve oscillates with the discrete 20 mm SiPM pitch.
- **Open — Gerardo:** Is T4/T20 a cut on total N_{pe} (as implemented) or a SUM4 trigger efficiency?

F5: SUM4 L/R timing resolution — EJ-204

$\sigma_t = \sigma(\Delta T_{LR})/\sqrt{2}$; jitter 20 ps/hit (Stage 1); no time-walk or readout jitter

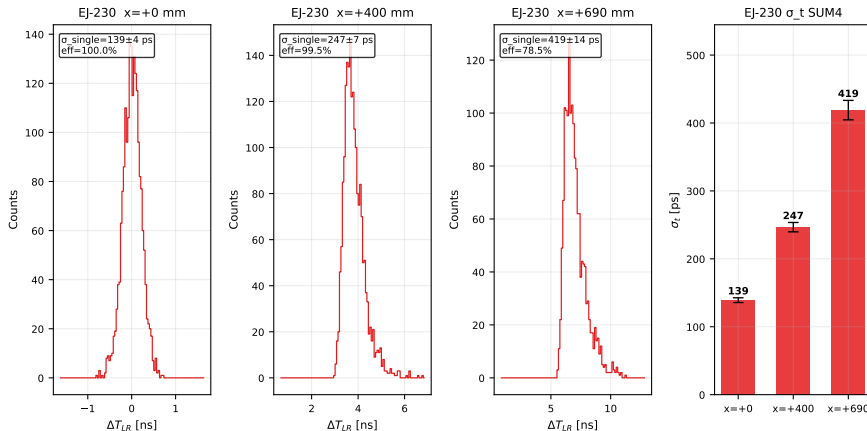
F5 — SUM4 L/R timing resolution (EJ-204)
 $\sigma_t = \sigma(\Delta T_{LR})/\sqrt{2}$ (intrinsic + 20 ps/hit jitter, no additional readout jitter)



F5: SUM4 L/R timing resolution — EJ-230

$\sigma_t = \sigma(\Delta T_{LR})/\sqrt{2}$; same Stage 1 caveats — stronger positional degradation than EJ-204

F5 — SUM4 L/R timing resolution (EJ-230)
 $\sigma_t = \sigma(\Delta T_{LR})/\sqrt{2}$ (intrinsic + 20 ps/hit jitter, no additional readout jitter)



Material	$x = 0 \text{ mm}$ $\sigma_t \text{ [ps]}$	$x = +400 \text{ mm}$ $\sigma_t \text{ [ps]}$	$x = +690 \text{ mm}$ $\sigma_t \text{ [ps]}$
EJ-204	141 ± 4	206 ± 6	365 ± 10
EJ-230	139 ± 4	247 ± 7	419 ± 14

Key observations:

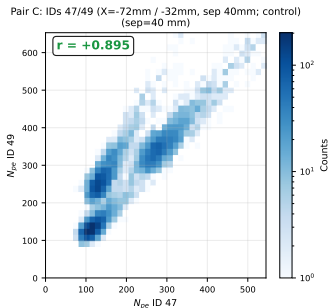
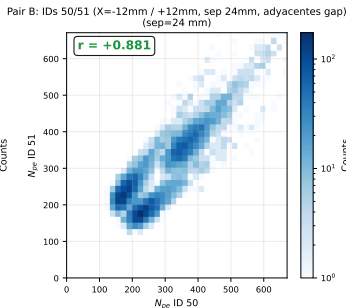
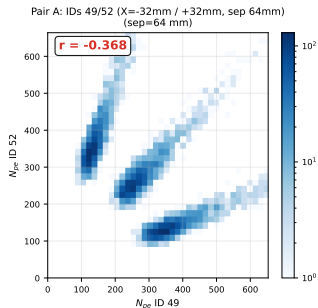
- Central resolution **equivalent**: $141 \approx 139 \text{ ps}$
- EJ-204 $\sim 15\%$ better at the edge (365 vs 419 ps)
- EJ-230 degrades faster with position
(+201% vs +159% from $x = 0$ to $x = 690$)
- Fitted τ_f : arrival-time profile (emission $\otimes$ propagation)
not datasheet constants
(EJ-204: $\tau_r^{(0)} = 0.7 \text{ ns}$, $\tau_d^{(0)} = 1.8 \text{ ns}$)

Caveats — Stage 1 (no real electronics):

- Jitter 20 ps/hit: possible double-count (L+R)
- No time-walk correction (ToT)
- No readout jitter applied
- SUM4 topology: pending Gerardo
- Recovering emission constants
requires deconvolution — Stage 2

F6: Top SiPM redundancy — N_{pe} correlation between neighbors

F6 — Top SiPM redundancy: N_{pe} correlation between neighbouring pairs
Pair A (49/52, symmetric): $r = -0.37$ complementary | Pair B (50/51): $r = +0.88$ | Pair C (47/49, same side): $r = +0.89$ redundant



Pair A (IDs 49/52)
sep = 64 mm, opposite sides
 $r = -0.37$ — complementary
(light splits between sides)

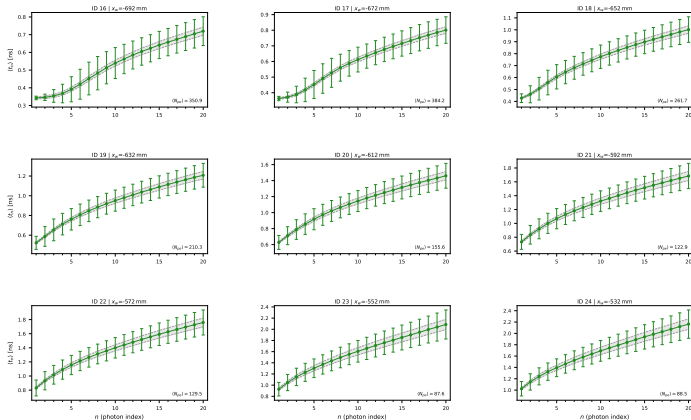
Pair B (IDs 50/51)
sep = 24 mm, across gap
 $r = +0.88$ — **redundant**
(same light source)

Pair C (IDs 47/49)
sep = 40 mm, same side
 $r = +0.90$ — **control**
(local correlation expected)

F7: Order-statistic arrival time $\langle t_n \rangle$ per Top channel — $x = -690$ mm

Statistical floor $\sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.8$ ns, EJ-204 datasheet); gap = optical path dispersion

F7 — Order-statistic arrival time (t_n) per Top channel | $x_{gap} = -690$ mm (EJ-204 EndTop)
Error bars = RMS (event-to-event). Dashed band: stat. floor $\sqrt{n} \tau_d / \langle N_{pe} \rangle$; $\tau_d = 1.8$ ns



What the axes show:

- x-axis: n = index of the n -th detected photon (order statistic)
- y-axis: $\langle t_n \rangle$ [ns] — mean arrival time per event
- Error bars: RMS over events
- Band: statistical floor
 $\sigma_{stat}(t_n) = \sqrt{n} \tau_d / \langle N_{pe} \rangle$

Physics:

RMS — σ_{floor} isolates **optical path dispersion** — the same spread that broadens σ_t and shifts v_{eff}^{onset} vs nominal.

At $x = -690$: cluster uses IDs 16-24 ($x_W = -692 \rightarrow -532$ mm) — truncated at the left edge; no channels further left.

F7 & F2 consistent: dispersion EXISTS here (RMS > floor) but is subdominant to Poisson statistics in the SUM4 σ_t over the accessible N_{pe} range ($b \approx 0$ in F2).

Main results:

- 1 **EJ-204** $\approx$ **EJ-230** at center (141 vs 139 ps); EJ-204 **better at edges** (365 vs 419 ps at $x = 690$)
- 2 F1 component at ~ 4.7 ns at $x = 690$ = **geometric ToF** from far end (QA-1c, $\delta = -5\%$). Not slow scintillation.
- 3 $v_{\text{eff}}^{\text{nominal}} = 277$ mm/ns; onset gives $v_{\text{eff}}^{\text{onset}} \approx 292$ mm/ns ($+5\%$) — fastest optical paths set the onset
- 4 F7: $\langle t_n \rangle$ RMS exceeds the statistical floor (order-statistics limit) — quantifies optical path dispersion; same physics as σ_t degradation toward edges
- 5 Top: robust > 20 PE; gap-adjacent SiPMs highly correlated ($r \approx 0.88$)

Open — Gerardo confirmation:

- ☐ SUM4 topology: winner-takes-first vs cluster sum
- ☐ T4/T20: N_{pe} cut or trigger efficiency
- ☐ Jitter 20 ps: single or double-count (L+R)
- ☐ F3: SiPM impacts vs volume trajectories
- ☐ Pair A ($r = -0.37$): geometric or reflections
- ☐ EJ-230 tail at $x = 0$: slow mode or reflections?

Next steps:

- EJ-230 EndTop dataset (Top response)
- Stage 2: time-walk, readout jitter, real electronics
- Deconvolution: emission constants from arrival profiles